

2025 Lessons for 2026

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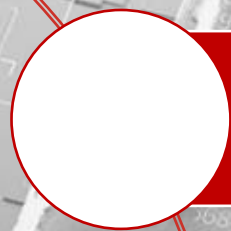
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{**elysium**security}
cyber protection & response

Welcome to this presentation on the 2025 Cyber Security lessons for 2026. We will discuss the incidents, attack patterns, technologies and strategic lessons that should shape cyber risk preparation in 2026.

1. CONTEXT: 2025 THREAT LANDSCAPE OVERVIEW
2. 2025: KEY INCIDENTS, TRENDS, AND TAKEAWAYS
3. 2026: EMERGING RISKS AND ATTACK VECTORS
4. NEXT: STRATEGIC RECOMMENDATIONS



CONTEXT



2025



2026



NEXT

\$10.5T

Annual Cybercrime cost globally - 3rd world GDP
(Cybersecurity Ventures)

2026 Prediction: **WORSE**

\$4B

Crypto Stolen
(beincrypto/Peckshield)

2026 Prediction: **WORSE**

\$4.44M

Average cost of a data breach - 9% decrease
(IBM)

2026 Prediction: **SAME**

4.8M

Unfilled Cybersecurity jobs worldwide - 19% increase
(ISC2)

2026 Prediction: **SAME**

16B+

Credentials leaked on the dark web - Some duplicates
(Cybernews)

2026 Prediction: **WORSE**

53%

of breaches used stolen credentials
(Verizon)

2026 Prediction: **SAME**

29 min

Average attacker breakout time - 65% speed increase
(CrowdStrike)

2026 Prediction: **WORSE**

36K/s

Malicious scans per second detected - 17% increase
(Fortinet)

2026 Prediction: **WORSE**



GEOPOLITICAL CYBER ESCALATION

Russia, China, North Korea and Iran intensified state-sponsored operations. Chinese attacks on Taiwan doubled to 2.4M daily attempts. Russia's attacks on Ukraine surged 70%.



RANSOMWARE AS A SERVICE

Law enforcement dismantled major groups, but the ecosystem became more decentralized. Over 85 unique ransomware groups operated in H1 2025, with RaaS lowering barriers to entry: Teenage hackers with enterprise-grade tools



SUPPLY CHAIN RISK

Open-Source & commercial software poisoning, SaaS integration and OAuth tokens targeted, 3rd party are targets of choice and sometimes the easiest prey.



AI-AUGMENTED ATTACKS

AI enhanced both attack and defense. Deepfake-as-a-Service platforms emerged. Phishing AI enhanced, automated Vulnerability scanning, Social engineering became harder to detect.



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- Fragmentation: 45 new groups merged -> A record 85 total
- Increased attack surface: 3rd party and Cloud
- Increased speed of attack: Less time to react
- Payment rates drop: Companies less willing to pay
- Double Extortion: Now standard to combine Encryption + Theft
- Triple Extortion increase: Encryption + Theft + Harassing Victim



Ransomware ecosystem is more distributed and resilient. Incident response plans must account for multi-vector extortion scenarios.

4x Increase in 5 years

30% of all breaches in 2025 alone, with supply chain attacks now averaging 26 incidents per month. Third-party vendors account for 80%+ of stolen healthcare records.

Data Mapping Gaps

Organizations have murky understanding of where data is, which vendors have access, and data volumes. Strategic third-party risk management programs are essential.

Vendor Security Blindspot

Even well-resourced organizations fell victim when a shared vendor was compromised. The UK retail campaign showed how one third-party can bring down an entire sector.

SaaS Integration Risks

The Salesforce/Drift breach exposed OAuth tokens and refresh tokens, allowing indirect access to customer environments. Up to 1.5B CRM records potentially impacted.



"We're moving into a world of untrusted systems."
(Bruce Schneier)

Review SaaS trust path as you would an internet facing asset

- Credential abuse, infostealers and help-desk social engineering increasingly converged
- Most identity attacks still exploited simple weaknesses: reused passwords, weak MFA paths and token abuse
- Phishing bypassing MFA through more realistic fake MFA site and response speed



Attackers are increasingly signing-in, not only breaking in. Identity is the new perimeter.



AI-POWERED ATTACKS

- 42% rise in phishing attacks fueled by GenAI content
- AI voice synthesis used for vishing & CEO fraud
- Real-time deepfake video calls for impersonation scams
- Automated vulnerability scanning & exploit generation at scale
- AI-powered malware that adapts to evade detection



AI-POWERED DEFENSE

- AI agents automating SOC triage: reducing alert fatigue by 70%+
- Behavioral analytics detecting anomalous access in real-time
- AI-driven threat intelligence correlating indicators at scale
- Automated incident response & containment playbooks
- **Only 37% have safeguards to assess AI tools before deployment**



AI was an amplifier in 2025: real, important and powerful, but it was not a substitute for fundamentals.

Secure and Govern AI usage within your organizations.

FROM

Prevent all breaches

Perimeter-focused defense

Annual penetration tests

Compliance-driven security

IT team owns cybersecurity



TOWARDS

Assume breach – focus on recovery

Zero-trust architecture (back again!)

Continuous exposure management

Risk-driven, board-level priority

Organization-wide responsibility



“The focus is shifting from achieving security systems to prioritizing defensible, resilient and recoverable systems”

(Dark Reading 2026 – Rob Wright)



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SHADOW AGENT

Gartner forecasts 40% of enterprise apps will have task-specific AI agents by 2026.

And only 6% have advanced AI security strategies today.

'Shadow agent' risk is the new shadow IT.



AGENTIC AI ATTACKS

AI agents will automate the entire attack lifecycle:

Recon
Scanning
Exploitation
Persistence

Google forecasts adversaries shifting to these type of attacks as norm.



PROMPT INJECTION

Google identifies prompt injection as a critical growing threat.

Manipulating AI systems to bypass security protocols and follow attacker commands.



Define and deploy AI Governance frameworks.
Assess every AI tool before deployment.
Build an "Agentic SOC": AI agents defending AI attacks



- **CEO Doppelgänger:** Fake real-time leader videos/audio/emails
- **Machine Identity Explosion:** There are more Non-Human identities than human identities already today
- **MFA is no longer enough:** Session hijacking, token theft, SIM-Swapping, etc.



Review your Identity verification process
Identity security must evolve to continuous, behavioural and adaptive authentication.

Cloud-Native Ransomware

Attackers increasingly target cloud environments: exploiting misconfigurations, cloud APIs, and backup systems.
I.e.: AWS environments already targeted by Crimson Collective for extortion.

AI Stack Supply Chain

Supply chain attacks will extend into AI stacks: ML models, training data, and AI pipeline dependencies become prime targets.
Vendor concentration amplifies systemic risk.

Hypervisor Attacks

As guest-level security improves, attackers focus on hypervisors and cloud control planes.
Compromising one hypervisor means controlling many systems at once.



Cloud security posture management needed
AI supply chain audits whenever possible
Importance of immutable backups and air-gapped recovery

CHINA – Critical infra compromise

Volt Typhoon & Salt Typhoon campaigns continue. Focus on semiconductor tech, Taiwan, and critical infrastructure pre-positioning for potential future conflict. 2025's ToolShell showed increasing sophistication.

RUSSIA – Hybrid Warfare

Blending ransomware with influence operations. AI-driven synthetic media campaigns to manipulate public perception. NATO-aligned countries remain primary targets.

IRAN – No Limit

As Iran is at war with the US and Israel and is seen physically targeting a very wide range of targets/countries, what they will do in the cyber space might be broader than they ever did before.

Crime and State - Convergence

Lines between state-sponsored actors and cybercriminals continue blurring. Storm-2603 is 'hybrid': deploying ransomware while believed to be state-linked. State actors use criminal tools; criminals get state-level resources.

2026 Events – Business as usual

Milan-Cortina 2026 Olympics was identified as a prime cyber target. Election cycles across multiple countries will drive influence operations at scale.



Invest in a Counter Threat Intelligence service, or/and actively monitor external attacks trends.
Expect your company or one of your vendor to be attacked.

PROBLEM STATEMENT

Harvest now, Decrypt later.

Adversaries are already collecting encrypted data today with the expectation that future quantum computers will break current encryption standards. This means data stolen today could be exposed in the future.

- Conduct a cryptographic inventory
- Assess which data requires long-term confidentiality
- Begin testing NIST post-quantum algorithms
- Build crypto-agility: Algorithm rotation without impact
- EU Milestone: National PQC roadmaps by 2026 for gov. & critical infra



Prepare Now.
Leg work will be useful today.



EU AI Act Enforcement

The world's first comprehensive AI law hits full enforcement on 2 August 2026 for high-risk AI systems, with fines up to €35 million or 7% of global turnover, exceeding even GDPR penalties

NIS2 And DORA

Two EU regulations now mandate cybersecurity risk management, supply chain security, and 24-hour incident reporting across 18 critical sectors and all financial services, with enforcement live and no grace period left.

Executive Liability

Cybersecurity failures are now personally actionable — executives face suspension, fines, and criminal charges under NIS2, DORA, SEC rules, and precedents set by the SolarWinds and Uber CISO prosecutions.

Data Sovereignty

The EU, China, India, and the U.S. are all mandating where data must live and whose laws govern it: forcing organizations to architect for a fragmented world where cloud strategy is now a legal and geopolitical decision.



Audit every AI System you deploy now
Know where your data lives and whose laws govern it
Cybersecurity is someone's personal liability (you?)

2026 SCOPE	THREAT VECTOR EXAMPLES	PROBABILITY 2026	POTENTIAL IMPACT	SPEED OF ATTACK
AI ARM RACE ACCELERATION	Agentic AI attacks	Emerging → Very High	Unknown scale	Seconds
	AI-powered phishing / BEC	Very High	High (credentials, access)	Minutes
IDENTITY CRISIS DEEPEN	Identity / credential theft	Very High	Critical (full breach access)	Seconds
RANSOMWARE AND SUPPLY CHAIN CONVERGENCE	Ransomware + data extortion	Very High (+40% vs 2024)	Catastrophic (\$5M+ avg)	Days -> Hours
	Supply chain compromise	High (doubled YoY)	Systemic / cascading	Weeks -> Days
NATION-STATE OPERATIONS INTENSIFY	Nation-state espionage	High (geopolitical)	Critical (IP + infra)	Months (silent)
	OT/ICS disruption	Medium-High	Societal (infra shutdown)	Variable
REGULATION, SOVEREIGNTY & LIABILITY	AI Standards and Laws	Low-Medium	Executives and companies' liabilities (Prison/Fines)	N/A
QUANTUM CLOCK IS TICKING	Quantum / HNDL	Low-now / High-future	Catastrophic (retroactive)	Unknown

KEYCODE	VERY HIGH	HIGH	MEDIUM	LOW
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1. **AI Governance and Usage:** Assess every AI tool before deployment and managed shadow agent. Leverage AI to your advantage!
2. **Identity-First Security:** Behavioral authentication, Machine identity governance and passwordless MFA in a Zero-Trust world.
3. **Regulation:** Ensure you are aware of existing and upcoming regulations, laws and standards
4. **Resilience and Recovery:** Ensure your DR, BCP and IR plans are up to date with today's technologies and challenges
5. **Supply Chain Resilience:** Continuous vendor monitoring, sovereignty assessments and AI stack audits
6. **Crypto Agility:** Start your cryptographic Inventory and build for algorithm rotation



PHASE 1

- Assess AI tool deployment and governance gaps
- Conduct a Data discovery/inventory
- Audit 3rd Party access and Security controls
- Review Incident Response plan readiness
- *Conduct cryptographic inventory*

PHASE 2

- Deploy/expand behavioral analytics
- Implement continuous vendor monitoring
- Launch AI security assessment process
- Conduct board-level tabletop exercise
- Test enhanced cyber security DR process

PHASE 3

- Operationalize AI in SOC triage
- Define identity-first architecture roadmap
- Establish supply chain SBOM requirements
- Publish updated risk metrics for board
- *Begin PQC algorithm testing*

**AI is a double-edge sword**

Govern AI rigorously on both offense and defense

**Identity is the new perimeter**

Invest in behavioral, adaptive, continuous authentication

**Regulation is accelerating**

Executive liability for AI governance failures is coming

**Resilience > prevention**

Assume breach, focus on detection, response, and recovery

**Supply chain risk is systemic**

Your security is only as strong as your weakest vendor

**Start Quantum preparation now**

Harvest-now-decrypt-later is already happening

THANK YOU!

Any Questions?



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